Midas_V2 — Multi■Date Testing Sequence (Validated) — v0.3.4 candidate

This note records exactly what we ran today, why earlier results looked off, how we fixed it, and the *validated* results for Scenario **D** with a fresh universe per day.

What changed (so today’s results are genuine)

- **topgappers.py** now **writes a fresh universe** every run (no `--no-write`), and tolerates extra args via `parse_known_args()`.

- **run_range_and_summarize.py** was fixed to use `--scenario` and to stop passing unsupported flags to `run_day_simple.py`.

- Verified on **2025■08■06**: `Wrote 94 symbols -> data\samples\universe_sample.txt` and backtest ran clean.

Commands we executed (Scenario D)

Short range (sanity check)

```powershell

python scripts\run_range_and_summarize.py --start 2025-08-05 --end 2025-08-10 --scenario D

python scripts\show_latest_range.py --root out\auto --scenario D

python scripts\analyze_range_explained.py --csv out\auto\range_summary_20250805_20250810_D.csv
```

Single-day verification (universe written & used)

```powershell

python scripts\run_day_simple.py --date 2025-08-06 --scenario D
```

observed: Wrote 94 symbols -> data\samples\universe_sample.txt

```

### Long range (validated)

```powershell

python scripts\run_range_and_summarize.py --start 2025-08-05 --end 2025-08-31 --scenario D

python scripts\show_latest_range.py --root out\auto --scenario D

python scripts\analyze_range_explained.py --csv out\auto\range_summary_20250805_20250831_D.csv
```

Results (Scenario D, 2025■08■05 → 2025■08■31)

- **Total trades:** 79

- **Wins / Losses:** 50 / 29

- **Overall Win Rate:** **63.29%**

- **Total PnL:** **+167.57**
```

- **Days with trades / zero■trade days:** 3 / 13
- **Best day:** 2025■08■06 (PnL +174.33, Trades 38, WR 65.79%)
- **Worst day:** 2025■08■05 (PnL –19.39, Trades 3, WR 0.00%)

Interpretation: D exceeds the 55% WR target. Good candidate baseline. Many zero■trade days → we can broaden universe or ease guards later *after* baseline is locked.

> Note: Earlier **B** results were produced before the universe fix and may reflect a STTK■only universe. Re■run B after the fix to get a fair baseline.

Next steps

1) Run **D** across more dates (e.g., 2025■09■01 → 2025■09■30) to increase sample size.
2) Re■run **B** (fresh universe) for a true baseline comparison.
3) Trial **E (dip■reclaim)** against D to evaluate added entries vs WR.
4) Only after baselines: consider 1■second entries and multiple trades per ticker.

Version tagging

Proposed tag: `v0.3.4-rangefix`

Two lines:

```powershell
git add -A; git commit -m "Docs: validated multi-date testing; D >60% WR with daily universe; range runner & gappers fixed"
git tag -a v0.3.4-rangefix -m "Milestone: daily-universe validation; Scenario D ~63% WR (Aug '25)"; git push; git push --tags
```

Appendix — Analyzer outputs (verbatim)

D (Validated) — range_summary_20250805_20250831_D.csv

```text
------------------------------------------------------------------------
Range Analysis — range_summary_20250805_20250831_D.csv
------------------------------------------------------------------------
Overview
Rows : 16 (days in range)
Days with trades : 3
Days with zero trades : 13
Totals
Total Trades : 79
Wins / Losses : 50 / 29
Overall Win Rate : 63.29% (wins / (wins+losses))
```

Total PnL : 167.57 (sum of daily PnL)
Daily Quality (on trade days)
Avg Daily Win Rate : 43.86%
Median Daily Win Rate : 65.79%
Best Day : 2025-08-06 PnL +174.33 Trades 38 WR 65.79%
Worst Day : 2025-08-05 PnL -19.39 Trades 3 WR 0.00%

Interpretation
✔ Baseline WR is in the target zone (≥55%). Continue expanding the sample.
Note: Many zero■trade days — consider broadening the universe or easing filters later (after baseline).

Next Actions
1) Run another range to increase sample size (same scenario).
2) If WR <55%, compare with Scenario D_strict (tighter early guard) or E_dip (reclaim entries).
3) Only after baseline is validated, consider 1■sec entries or multiple trades per ticker.

```

### B (Pre-fix, stale universe) — range_summary_20250805_20250831_B.csv

```text
----------------------------------------------------------------------
Range Analysis — range_summary_20250805_20250831_B.csv
----------------------------------------------------------------------
Overview
Rows : 16 (days in range)
Days with trades : 7
Days with zero trades : 9
Totals
Total Trades : 91
Wins / Losses : 45 / 46
Overall Win Rate : 49.45% (wins / (wins+losses))
Total PnL : -92.44 (sum of daily PnL)
Daily Quality (on trade days)
Avg Daily Win Rate : 41.78%
Median Daily Win Rate : 44.44%
Best Day : 2025-08-05 PnL +76.78 Trades 27 WR 66.67%
Worst Day : 2025-08-08 PnL -79.96 Trades 20 WR 50.00%
----------------------------------------------------------------------
Interpretation
✘ WR < 50%. Prioritize guard tuning before expanding sample size.

Note: Many zero■trade days — consider broadening the universe or easing filters later (after baseline).

----------------------------------------------------------------------

Next Actions

1) Run another range to increase sample size (same scenario).

2) If WR <55%, compare with Scenario D_strict (tighter early guard) or E_dip (reclaim entries).

3) Only after baseline is validated, consider 1■sec entries or multiple trades per ticker.

----------------------------------------------------------------------

```

Appendix — Scenario **D (strict)** explained, end■to■end

Goal. Scenario **D** is our stricter momentum profile designed to avoid early■session chop and prioritize higher win rate. It keeps confirmations tight and limits entries early in the day with a short **gate**.

A) Data & universe (inputs)

1. **Daily universe** comes from `scripts/topgappers.py` (now writes fresh each run).

- Source: Polygon grouped bars (today open vs yesterday close).

- Defaults: `--min-price 1`, `--max-price 20`, `--min-gap 5`, `--top 50`.

- Output: `data/samples/universe_sample.txt` (one symbol per line).

2. **Minute data** is fetched per symbol/day by `scripts/fetch_minutes_polygon.py --session rth` and cached under `data/samples/*.csv`.

> Why you saw zero■trade days earlier: before the fix, the universe was stale (often STTK■only). Now it refreshes daily; zero■trade days simply mean no setups met D's stricter conditions that day.

B) Current default parameters (from run logs)

```
min_pm_vol = 30000 # premarket liquidity floor
ema_confirm = True # require 9■EMA alignment
vwap_confirm = True # require VWAP alignment
ema_period = 9
macd_confirm = True # MACD must be supportive
rise_bars = 2 # need 2 rising MACD bars
gate_minutes = 5 # wait the first 5 minutes after the open
tp_pct = 2.0 # take■profit at +2.0%
sl_pct = 2.5 # stop■loss at –2.5%
dip_reclaim = False # (off in D) dip■reclaim entries disabled
reclaim_pmh = False # (off) breakout over pre■market high not required
min_reclaim_pct = 0.5 # if reclaim logic is used by the engine, min reclaim size (bps)
reclaim_ref = "ema" # reclaim measured against EMA baseline
```

These came from the engine banner during your runs on 2025■08■06 and others.
```

C) Confirmation stack (what must be true)

- **EMA(9) & VWAP confirm**: price action aligns above/with these baselines (helps filter weak pops).

- **MACD confirm (rise_bars=2)**: histogram rising for ≥2 bars, momentum direction supportive.

- **Premarket volume ≥ 30k**: thin issues are filtered out.

D) Timing guard (gate)

- **gate_minutes = 5**: No entries allowed until 9:35 ET. This cuts out the noisiest 5 minutes.

E) Entry logic (how a trade is allowed)

- The engine requires the **confirmation stack** first.

- With **dip_reclaim=False** and **reclaim_pmh=False**, D focuses on **clean momentum continuation** with EMA/VWAP conformity and MACD support.

- Where the engine uses reclaim sizing, **min_reclaim_pct=0.5**% and **reclaim_ref='ema'** provide a minimum move against the EMA to qualify the trigger.

- Result: fewer, cleaner entries than B; typically higher WR, lower frequency.

F) Exits & risk

- **Hard TP/SL**: +2.0% TP, −2.5% SL.

- **EOD flatten**: positions closed by end of day (no overnights).

- Engine logging writes `results_YYYY-MM-DD.csv` per day and scenario with PnL outcomes.

G) Why D can show many zero■trade days

- Conditions are **strict**: EMA+VWAP+MACD alignment plus gate and liquidity floor.

- Some days simply have no compliant setups in the price/gap band. That’s acceptable for a high■WR baseline.

H) How D differs from B (baseline)

- **D is stricter**: shorter gate (5 vs 10), identical TP/SL, same EMA/VWAP+MACD confirmations, dip■reclaim disabled.

- **Effect**: fewer early whipsaws, fewer trades overall, **higher WR** in your August sample.

I) Recommended usage

- Use **D** as your safer baseline while building sample size and validating.

- Once D is stable (≥55% WR over many dates), compare against **E (dip■reclaim)** to potentially add entries without sacrificing WR.

J) Quick commands (recap)

```powershell

# Run long D range
python scripts\run_range_and_summarize.py --start 2025-08-05 --end 2025-08-31 --scenario D

# Show latest D summary
python scripts\show_latest_range.py --root out\auto --scenario D

# Explain totals (D)
python scripts\analyze_range_explained.py --csv
out\auto\range_summary_20250805_20250831_D.csv
```

K) Troubleshooting checklist

- Universe shows “Wrote N symbols”: if **N=0**, check Polygon key / market holiday.

- Too many zero■trade days? Confirm `min_pm_vol`, price band, and MACD `rise_bars`.

- Unexpected arg errors? You’re already on the hardened `topgappers.py` (parse_known_args, `--out`).
Re■run `Docs/refresh_docs.py` if scripts list looks stale.